**Paper 2**

| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| **1** | **a** | | | |
| $\bullet^1$ | ss | find gradient of AB | $\bullet^1$ $m_{AB}=1$ | |
| $\bullet^2$ | pd | find perpendicular gradient | $\bullet^2$ $m_{perp}=-1$ stated or implied by $\bullet^4$ | |
| $\bullet^3$ | pd | find midpoint of AB | $\bullet^3$ $(4,1)$ stated or implied by $\bullet^4$ | |
| $\bullet^4$ | pd | obtain equation | $\bullet^4$ $y-1=-1(x-4)$ | **4** |

**Notes:**

1. $\bullet^4$ is only available as a consequence of using a perpendicular gradient **and** a midpoint.
2. The gradient must appear in simplified form at $\bullet^4$ stage for $\bullet^4$ to be awarded.

**Commonly Observed Responses:**

**Candidate A**

$m_{AB}=-1$ $\bullet^1$ X

$m_{perp}=1$ $\bullet^2$ ✓

$(4,1)$ $\bullet^3$ ✓

$y-1=1(x-4) \Rightarrow y=x-3$ $\bullet^4$ ✓

Leading to part (b)

$y-x=-3$
$y+2x=6$ $\bullet^5$ ✓

$(3,0)$ $\bullet^6$ ✓

$\bullet^7$ and $\bullet^8$ are not available as $A=T=(3,0)$



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| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| 1 | b | | | |
| $\bullet^5$ ss | | know to solve simultaneously | $\bullet^5$ $y+2x=6$, $y+x=5$ | |
| $\bullet^6$ pd | | solve correctly for $x$ and $y$ | $\bullet^6$ $x=1,\ y=4$ | 2 |

**Commonly Observed Responses:**

**Candidate B**

Part (a) $y-1=-1(x-4)$ $\bullet^4$ ✓

$y=-x+3$ error

Part (b) $y+2x=6$ and $y+x=3$ $\bullet^5$ ✓

$x=3,\ y=0$ $\bullet^6$ ✗ correct strategy used, pd mark not available

| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| 1 | c | | | |
| $\bullet^7$ ss | | know and use $m=\tan\theta$ | $\bullet^7$ $\tan\theta=-2$ | |
| $\bullet^8$ pd | | calculate angle | $\bullet^8$ $116\cdot6^o$ accept $117^0$ or $2\cdot03$ radians | 2 |

**Commonly Observed Responses:**

**Candidate C**

$m_{AT}=-\frac{1}{2}$

base angle $=26\cdot6^o$ $\bullet^7$ ✗

$\Rightarrow$ angle $=90+26\cdot6=116\cdot6^o$ $\bullet^8$ ✗

**Candidate D**

$m_{AT}=2$ $\bullet^7$ ✗

angle $=\tan^{-1}(2)=63\cdot4^o$ $\bullet^8$ ✗

**Candidate E:**

Part (a)

$m_{AB}=\frac{2-0}{5-3}=\frac{2}{8}=\frac{1}{4}$ $\bullet^1$ ✗

$m_{perp}=-4$ $\bullet^2$ ✗

Midpoint of AB (4, 1) $\bullet^3$ ✓

$y-1=-4(x-1)$ $\bullet^4$ ✗

$y+4x-5$

Part (b)

$y+4x-5=0$, $y+2x+6=0$ $\bullet^5$ ✗ $\Rightarrow$ $y+2x=-6$, $y+4x=-5$ $\bullet^6$ ✗

$\Rightarrow 2x=1,\ x=\frac{1}{2},\ y=-7$

$\bullet^5$ is a strategy mark. The correct strategy is to solve the **given equation** with the equation from part (a) simultaneously. $\bullet^5$ is not awarded as the given equation has not been used.

The equation obtained at stage $\bullet^4$, has been rearranged incorrectly in part (b). The next pd mark, $\bullet^6$, is therefore not awarded.



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| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| **2** | | | | |
| $\bullet^1$ | ss | know to and differentiate | $\bullet^1$ $4x^3-6x^2$ | |
| $\bullet^2$ | ic | find gradient | $\bullet^2$ 8 | |
| $\bullet^3$ | pd | find $y$-coordinate | $\bullet^3$ 5 | |
| $\bullet^4$ | ic | state equation of tangent | $\bullet^4$ $y-5=8(x-2)$ | **4** |

**Notes:**

1. $\bullet^4$ is only available if an attempt has been made to find the gradient from differentiation **and** calculating the *y*-coordinate by substitution into the original equation.

**Commonly Observed Responses:**

**Candidate A**

$\bullet^1$ ✓ $\bullet^2$ ✓ $\bullet^3$ ✓

using $y=mx+c$

$x=2,\ y=5,\ m=8$

$\Rightarrow 5=8\times 2+c$

$\Rightarrow c=-11$ $\bullet^4$ ✓

$y=8x-11$



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| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| 3 | a | | | |
| $\bullet^1$ ic interpret notation<br>$\bullet^2$ pd a correct expression | | | $\bullet^1$ $f(x+3)$ stated or implied by $\bullet^2$<br>$\bullet^2$ $=(x+3)(x+2)+q$<br>**OR**<br>$=(x+3)^2-(x+3)+q$<br>or equivalent | **2** |

**Notes:**

1. Special Case: $\bullet^1$ is for substituting $(x+3)$ for $x$ thus, treat $x+3(x+3-1)+q$ as bad form.

**Commonly Observed Responses:**

**Candidate A**

$f(g(x))=x+3(x+3-1)+q$ $\bullet^1$ ✓

$=x^2+5x+6+q$ $\bullet^2$ ✓ $\bullet^3$ ✓

**Candidate B**

$f(g(x))=x+3(x+3-1)+q$ $\bullet^1$ ✓

$=4x+6+q$ $\bullet^2$ X

**Candidate C**

$f(g(x))=x+3(x+3-1)+q$ $\bullet^1$ ✓

$=(x+3)^2-x+3+q$

$x^2+5x+6+q=0$ $\bullet^2$ ✓ $\bullet^3$ ✓

**Candidate D**

$f(g(x))=(x+3)(x+3-1)+q$ $\bullet^1$ ✓ $\bullet^2$ ✓

$=(x+3)^2-x+3+q$

$x^2+5x+12+q=0$ $\bullet^3$ X

**Candidate E:** using $g(f(x))$

part (a)

$g(f(x))=g(x(x-1)+q)$ $\bullet^1$ X

$=x(x-1)+q+3$ $\bullet^2$ ✓

Leading to …..

part (b)

$x^2-x+q+3=0$ $\bullet^3$ ✓ (eased)

$b^2-4ac=(-1)^2-4\times1\times(q+3)$ $\bullet^4$ ✓

$1-4q-12=0$ $\bullet^5$ ✓

$q=-\frac{11}{4}$ $\bullet^6$ ✓



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| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| 3 | b | | | |
| | | **Method 1** | **Method 1** | |
| $\bullet^3$ | pd | write in standard quadratic form | $\bullet^3$ $x^2 + 5x + 6 + q = 0$ | |
| $\bullet^4$ | ic | use discriminant | $\bullet^4$ $b^2 - 4ac = 5^2 - 4\times1\times(6+q)$ | |
| $\bullet^5$ | pd | simplify and equate to zero | $\bullet^5$ $\Rightarrow 25 - 24 - 4q = 0$ | |
| $\bullet^6$ | pd | find value of $q$ | $\bullet^6$ $q = \frac{1}{4}$ | **4** |
| | | **Method 2** | **Method 2** | |
| $\bullet^3$ | pd | write in standard quadratic form | $\bullet^3$ $x^2+5x+6+q=0$ | |
| $\bullet^4$ | ic | complete the square | $\bullet^4$ $\left(x+\frac{5}{2}\right)^2 - \frac{25}{4}+6+q=0$ | |
| $\bullet^5$ | pd | equate to zero | $\bullet^5$ $-\frac{25}{4}+6+q=0$ | |
| $\bullet^6$ | pd | find value of $q$ | $\bullet^6$ $q=\frac{1}{4}$ | |
| | | **Method 3** | **Method 3** | |
| $\bullet^3$ | pd | write in standard quadratic form | $\bullet^3$ $f(g(x)) = x^2+5x+6+q=0$ | |
| $\bullet^4$ | ic | geometric interpretation | $\bullet^4$ equal roots so touches $x$-axis at SP | |
| $\bullet^5$ | pd | differentiates to obtain $x$ | $\bullet^5$ $\Rightarrow \frac{dy}{dx} = 2x+5=0$<br>$x=-\frac{5}{2}$ | |
| $\bullet^6$ | pd | find value of $q$ | $\bullet^6$ $\frac{25}{4}-\frac{25}{2}+6+q=0$<br>$q=\frac{1}{4}$ | |

**Notes:**

2. Do not penalise the omission of ' $=0$ ' at $\bullet^3$.
3. In Method 1 $a=1,\ b=5,\ c=6+q$ is sufficient for $\bullet^3$.
4. Candidates who assume ' $=0$ ' and follow through to a correct value of $q$, $\bullet^6$ is still available. In Methods 1 and 2 ' $=0$ ' must appear at $\bullet^4$ or $\bullet^5$ for $\bullet^5$ to be awarded.
5. If the expression obtained at $\bullet^3$ is not a quadratic then $\bullet^3$, $\bullet^4$, $\bullet^5$ and $\bullet^6$ are not available.



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| Question | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|
| **Throughout this question treat coordinates written as components, and vice versa, as bad form.** | | | |
| **4 a** | | | |
| $\bullet^1$ pd states coordinates of C | | $\bullet^1$ C(11,12,6) | **2** |
| $\bullet^2$ pd states coordinates of D | | $\bullet^2$ $D(8,8,4)$ | |
| **Notes:** | | | |
| 1. Accept $x=11$, $y=12$ and $z=6$ for $\bullet^1$ and $x=8$, $y=8$ and $z=4$ for $\bullet^2$. | | | |
| 2. For candidates who write the coordinates as Cartesian triples and omit brackets in both cases, $\bullet^2$ is not available. | | | |
| **4 b** | | | |
| $\bullet^3$ pd finds $\overrightarrow{CB}$ | | $\bullet^3$ $\begin{pmatrix}0\\-8\\-4\end{pmatrix}$ | **2** |
| $\bullet^4$ pd finds $\overrightarrow{CD}$ | | $\bullet^4$ $\begin{pmatrix}-3\\-4\\-2\end{pmatrix}$ | |
| **Notes:** | | | |
| 3. For candidates who find both $\overrightarrow{BC}$ and $\overrightarrow{DC}$, only $\bullet^4$ is available (repeated error). | | | |
| **4 c** . | | | |
| $\bullet^5$ ss know to use scalar product applied to the correct angle | | $\bullet^5$ $\cos B\hat{C}D = \dfrac{\overrightarrow{CB}.\overrightarrow{CD}}{\lvert\overrightarrow{CB}\rvert\lvert\overrightarrow{CD}\rvert}$ stated or implied by $\bullet^9$ | **5** |
| $\bullet^6$ pd find scalar product | | $\bullet^6$ 40 | |
| $\bullet^7$ pd find $\lvert\overrightarrow{CB}\rvert$ | | $\bullet^7$ $\sqrt{80}$ | |
| $\bullet^8$ pd find $\lvert\overrightarrow{CD}\rvert$ | | $\bullet^8$ $\sqrt{29}$ | |
| $\bullet^9$ pd find angle | | $\bullet^9$ $33{\cdot}9^o$ | |
| **Notes:** | | | |
| 4. $\bullet^5$ is not available for candidates who choose to evaluate an incorrect angle. | | | |
| 5. $\bullet^9$ accept 33·8 to 34 degrees or 0·59 to 0·6 radians. | | | |
| 6. If candidates do not attempt $\bullet^9$, then $\bullet^5$ is only available if the formula quoted relates to the labelling in the question. | | | |
| 7. $\bullet^9$ is only available as a result of using a valid strategy. | | | |
| 8. $\bullet^5$ is not available for candidates who write $\cos\theta = \dfrac{40}{\sqrt{80}\times\sqrt{29}}$. Some reference to the labelling of the diagram **must** be made within their solution to part (c), to indicate they are attempting to find the correct angle. | | | |



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**Commonly Observed Responses:**

**Candidate A:** Cosine Rule

$\cos B\hat{C}D = \dfrac{CB^2 + CD^2 - BD^2}{2 \times CB \times CD}$ •5 ✓

$CB = \sqrt{80},\ CD = \sqrt{29},\ BD = \sqrt{29}$ •6 ✓ •7 ✓ •8 ✓

33·9° or 0·59 radians •9 ✓

**Candidate B**

$\cos B\hat{C}D = \dfrac{\overrightarrow{BC}.\overrightarrow{CD}}{|\overrightarrow{BC}| \times |\overrightarrow{CD}|}$ •5 X

$\overrightarrow{BC}.\overrightarrow{CD} = -40$ •6 ✓

$|\overrightarrow{BC}| = \sqrt{80}$ , $|\overrightarrow{CD}| = \sqrt{29}$ •7 ✓ •8 ✓

146·1° or 2·55 radians •9 ✓

**Candidate C**

$\cos B\hat{O}D = \dfrac{\overrightarrow{OB}.\overrightarrow{OD}}{|\overrightarrow{OB}| \times |\overrightarrow{OD}|}$ •5 X

$\overrightarrow{OB}.\overrightarrow{OD} = 128$ •6 ✓

$|\overrightarrow{OB}| = \sqrt{141}$ , $|\overrightarrow{OD}| = 12$ •7 ✓ •8 ✓

26·1° or 0·46 radians •9 ✓

**Candidate D**

$\cos C\hat{B}D = \dfrac{\overrightarrow{BC}.\overrightarrow{BD}}{|\overrightarrow{BC}| \times |\overrightarrow{BD}|}$ •5 X

$\overrightarrow{BC}.\overrightarrow{BD} = 40$ •6 ✓

$|\overrightarrow{BC}| = \sqrt{80}$ , $|\overrightarrow{BD}| = \sqrt{29}$ •7 ✓ •8 ✓

33·9° or 0·59 radians •9 ✓

**Candidate E**

$\cos B\hat{O}C = \dfrac{\overrightarrow{OB}.\overrightarrow{OC}}{|\overrightarrow{OB}| \times |\overrightarrow{OC}|}$ •5 X

$\overrightarrow{OB}.\overrightarrow{OC} = 181$ •6 ✓

$|\overrightarrow{OB}| = \sqrt{141}$ , $|\overrightarrow{OC}| = \sqrt{301}$ •7 ✓ •8 ✓

28·5° or 0·50 radians •9 ✓

**Candidate F**

$\cos B\hat{C}D = \dfrac{\overrightarrow{BC}.\overrightarrow{DC}}{|\overrightarrow{BC}| \times |\overrightarrow{DC}|}$ •5 ✓

this is an acceptable form for the scalar product.

| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| 5 | | | | |
| $\bullet^1$ | ss | start to integrate | $\bullet^1$ $\frac{1}{\frac{1}{2}}(.....)^{\frac{1}{2}}$ | |
| $\bullet^2$ | pd | complete integration | $\bullet^2$ $...\times\frac{1}{3}$ | |
| $\bullet^3$ | pd | process limits | $\bullet^3$ $\frac{2}{3}(3t+4)^{\frac{1}{2}}-\frac{2}{3}(3(4)+4)^{\frac{1}{2}}$ | |
| $\bullet^4$ | pd | start to solve equation | $\bullet^4$ $(3t+4)^{\frac{1}{2}}=7$ | |
| $\bullet^5$ | pd | solve for $t$ | $\bullet^5$ $t=15$ | 5 |

**Notes:**

1. $\bullet^3$ is awarded for correct substitution leading to $F(t)-F(4)$ where $F(x)$ is the candidates attempt
2. to integrate $(3x+4)^{-\frac{1}{2}}$ . For substituting into the original function $\bullet^3$ is unavailable.
3. $\bullet^5$ is only available as a consequence of squaring both sides of an equation.
4. The integral obtained must contain a non integer power for $\bullet^4$ and $\bullet^5$ to be available.
5. Do not penalise the inclusion of '$+c$'.
6. Incorrect expansion of $(...)^{-\frac{1}{2}}$ at stage $\bullet^1$, only $\bullet^3$ is available as follow through. Incorrect expansion of $(...)^{\frac{1}{2}}$ at stage $\bullet^4$, $\bullet^4$ and $\bullet^5$ are not available.

**Commonly Observed Responses:**

**Candidate A:** Forgetting the $\frac{1}{3}$

$\left[2(3x+4)^{\frac{1}{2}}\right]_4^t=2$ $\bullet^1$ ✓ $\bullet^2$ X

$\left(2(3t+4)^{\frac{1}{2}}\right)-\left(2(3(4)+4)^{\frac{1}{2}}\right)=2$ $\bullet^3$ ✓

$(3t+4)^{\frac{1}{2}}=5$ $\bullet^4$ ✓

$t=7$ $\bullet^5$ ✓

**Candidate B**

$\left[\frac{1}{6}(3x+4)^{\frac{1}{2}}\right]_4^t=2$ $\bullet^1$ X $\bullet^2$ ✓

$\left(\frac{1}{6}(3t+4)^{\frac{1}{2}}\right)-\left(\frac{1}{6}(3(4)+4)^{\frac{1}{2}}\right)=2$ $\bullet^3$ ✓

$(3t+4)^{\frac{1}{2}}=16$ $\bullet^4$ ✓

$t=84$ $\bullet^5$ ✓

**Candidate C**

$\left[\frac{(3x+4)^{\frac{1}{2}}}{\frac{1}{2}}\times 3\right]_4^t=2$ $\bullet^1$ ✓ $\bullet^2$ X

$\left[\frac{2}{3}(3x+4)^{\frac{1}{2}}\right]_4^t=2$

$\left[\frac{2}{3}(3t+4)^{\frac{1}{2}}\right]-\left[\frac{2}{3}(3(4)+4)^{\frac{1}{2}}\right]=2$ $\bullet^3$ X

$(3t+4)^{\frac{1}{2}}=7$ $\bullet^4$ ✓

$t=15$ $\bullet^5$ ✓

**Candidate D**

$\left[-\frac{3}{2}(3x+4)^{-\frac{3}{2}}\right]_4^t=2$ $\bullet^1$ X $\bullet^2$ X

$-\frac{3}{2}(3t+4)^{-\frac{3}{2}}-\left(-\frac{3}{2}\times 16^{-\frac{3}{2}}\right)=2$ $\bullet^3$ ✓

$(3t+4)^{\frac{3}{2}}=-\frac{192}{253}$ $\bullet^4$ ✓

decimal equivalent not accepted

$t=-1{\cdot}056$ $\bullet^5$ ✓



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| Question | | Generic Scheme | | Illustrative Scheme | Max Mark |
|---|---|---|---|---|---|
| **6** | | | | | |
| | $\bullet^1$ | ss | use correct double angle formula | $\bullet^1$ $\sin x - 2\left(1-2\sin^2 x\right)$ stated or implied by $\bullet^2$ | |
| | $\bullet^2$ | ss | arrange in standard quadratic form | $\bullet^2$ $4\sin^2 x + \sin x - 3 = 0$ | |
| | $\bullet^3$ | ss | start to solve | $\bullet^3$ $\left(4\sin x - 3\right)\left(\sin x + 1\right) = 0$ **OR** $\dfrac{-1 \pm \sqrt{(1)^2 - 4\times 4\times(-3)}}{2\times 4}$ | |
| | $\bullet^4$ | ic | reduce to equations in $\sin x$ only | $\bullet^4$ $\sin x = \dfrac{3}{4}$ and $\sin x = -1$ | |
| | $\bullet^5$ | pd | process to find solutions in given domain | $\bullet^5$ $0{\cdot}848$, $2{\cdot}29$ and $\dfrac{3\pi}{2}$ **OR** $\bullet^4$ $\sin x = \dfrac{3}{4}$ and $x = 0{\cdot}848,\ 2{\cdot}29$ $\bullet^5$ $\sin x = -1$, and $x = \dfrac{3\pi}{2}$ | **5** |

**Notes:**

1. $\bullet^1$ is not available for simply stating $\cos 2A = 1 - 2\sin^2 A$ with no further working.
2. In the event of $\cos^2 x - \sin^2 x$ or $2\cos^2 x - 1$ being substituted for $\cos 2x$, $\bullet^1$ cannot be awarded until the equation reduces to a quadratic in $\sin x$.
3. Substituting $1 - 2\sin^2 A$ or $1 - 2\sin^2 \alpha$ for $\cos 2\alpha$ at $\bullet^1$ stage should be treated as bad form provided the equation is written in terms of $x$ at stage $\bullet^2$. Otherwise, $\bullet^1$ is not available.
4. '$= 0$' must appear by $\bullet^3$ stage for $\bullet^2$ to be awarded. However , for candidates using the quadratic formula to solve the equation, '$= 0$' must appear at $\bullet^2$ stage for $\bullet^2$ to be awarded.
5. Candidates may express the equation obtained at $\bullet^2$ in the form $4s^2 + s - 3 = 0$ or $4x^2 + x - 3 = 0$. In these cases, award $\bullet^3$ for $\left(4s - 3\right)\left(s + 1\right) = 0$ or $\left(4x - 3\right)\left(x + 1\right) = 0$. However, $\bullet^4$ is only available if $\sin x$ appears explicitly at this stage.
6. $\bullet^4$ and $\bullet^5$ are only available as a consequence of solving a quadratic equation.
7. $\bullet^3$ , $\bullet^4$ and $\bullet^5$ are not available for any attempt to solve a quadratic written in the form $ax^2 + bx = c$ .
8. $\bullet^5$ is not available to candidates who work in degrees and do not convert their solutions into radian measure.
9. $\sin x + 4\sin^2 x - 3 = 0$ does not gain $\bullet^2$, unless $\bullet^3$ is awarded.



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**Commonly Observed Responses:**

**Candidate A**

$\bullet^1$ ✓ $\bullet^2$ ✓

$(4s-3)(s+1)=0$ $\bullet^3$ ✓

$s=\frac{3}{4},\ s=-1$ $\bullet^4$ ✗

$x=0{\cdot}848,\ 2{\cdot}29$ and $\frac{3\pi}{2}$ $\bullet^5$ ✓

**Candidate B**

$\bullet^1$ ✓

$4\sin^2 x+\sin x-3=0$ $\bullet^2$ ✓

$5\sin x-3=0$ $\bullet^3$ ✗

$\sin x=\frac{3}{5}$ $\bullet^4$ ✗

$x=0{\cdot}644,\ 2{\cdot}50$ $\bullet^5$ ✓

**Candidate C**

$\bullet^1$ ✓

$\sin x-2(1-2\sin^2 x)=1$

$\sin x-2+4\sin^2 x=1$

$4\sin^2 x+\sin x=3$ $\bullet^2$ ✓

$\sin x(4\sin x+1)=3$ $\bullet^3$ ✓

$\sin x=3,\ \ 4\sin x+1=3$ $\bullet^4$ ✗

no solution, $\sin x=\frac{1}{2}$

$x=\frac{\pi}{6},\ \frac{5\pi}{6}$ $\bullet^5$ ✓

**Candidate D**

$\bullet^1$ ✓

$\sin x-2(1-2\sin^2 x)=1$

$4\sin^2 x+\sin x-3=0$ $\bullet^2$ ✓

$4\sin^2 x+\sin x=3$

$\sin x(4\sin x+1)=3$ $\bullet^3$ ✓

$\sin x=3,\ \ 4\sin x+1=3$ $\bullet^4$ ✗

no solution, $\sin x=\frac{1}{2}$

$x=\frac{\pi}{6},\ \frac{5\pi}{6}$ $\bullet^5$ ✓

**Candidate E:** Reading $\cos 2x$ as $\cos^2 x$

$\sin x-2\cos^2 x=1$ $\bullet^1$ ✗

$\sin x-2(1-\sin^2 x)=1$

$2\sin^2 x+\sin x-3=0$ $\bullet^2$ ✓

$(2\sin x+3)(\sin x-1)=0$ $\bullet^3$ ✓

$\sin x=-\frac{3}{2},\ \ \sin x=1$ $\bullet^4$ ✓

no solution, $x=\frac{\pi}{2}$ $\bullet^5$ ✓

| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| **7** | **a** | | | |
| $\bullet^1$ | ss | know to and find intersection of line and curve | $\bullet^1$ $2x = 6x - x^2 \Rightarrow x = 0, x = 4$ | |
| $\bullet^2$ | ic | use " upper – lower" | $\bullet^2$ $\int\left(\left(6x - x^2\right) - 2x\right)dx$ | |
| $\bullet^3$ | pd | integrate | $\bullet^3$ $2x^2 - \frac{1}{3}x^3$ | |
| $\bullet^4$ | pd | substitute limits and evaluate | $\bullet^4$ $10\frac{2}{3}$ | |
| $\bullet^5$ | pd | evaluate area developed | $\bullet^5$ $10\frac{2}{3} \times 300 = 3200\,\text{m}^2$ | **5** |

**Notes:**

1. '0' appearing as the lower limit of the integral is sufficient evidence for $x = 0$ at $\bullet^1$ stage.
2. $\bullet^5$ is only available as a consequence of multiplying an **exact** answer at $\bullet^4$ stage.
3. The omission of $dx$ at $\bullet^2$ should not be penalised.
4. Where a candidate differentiates one or both terms $\bullet^3$, $\bullet^4$ and $\bullet^5$ are unavailable.
5. Do not penalise the inclusion of '$+ c$'.
6. Accept $\int\left(4x - x^2\right)dx$ for $\bullet^2$.



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**Commonly Observed Responses:**

**Candidate A**

$\int_0^4 \left(2x-\left(6x-x^2\right)\right)dx$ $\bullet^2$ X

$=\frac{1}{3}x^3-2x^2$ $\bullet^3$ ✓

$=-10\frac{2}{3}$ cannot be negative so $=10\frac{2}{3}$ $\bullet^4$ X however ….$=-10\frac{2}{3}$ so Area $=10\frac{2}{3}$ $\bullet^4$ ✓

Area = $3200\text{m}^2$ $\bullet^5$ ✓

**Candidate B**

$2x=6x-x^2 \Rightarrow x=0,\ 4$ $\bullet^1$ ✓

Shaded area

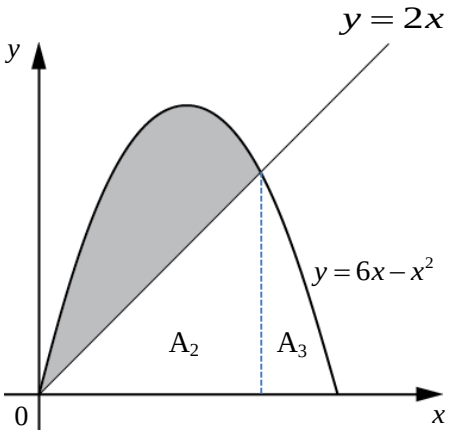
= area under parabola – ($A_2$ + $A_3$)

$=\int_0^6 \left(6x-x^2\right)dx-\left[A_2+\int_4^6 (6x-x^2)dx\right]$ $\bullet^2$ ✓

Stated or implied by $\bullet^4$

Area under parabola = 36, $A_2$ = 16 and $A_3 = \frac{28}{3}$ $\bullet^3$ ✓

Shaded area $=36-\left(16+\frac{28}{3}\right)=\frac{32}{3}$ $\bullet^4$ ✓




**Candidate C**

Part (a)

$x=0,\ x=6$ $\bullet^1$ X

$\int\left(\left(6x-x^2\right)-2x\right)dx$ $\bullet^2$ ✓

$\left[2x^2-\frac{1}{3}x^3\right]_0^6$ $\bullet^3$ ✓

$\left(2\times6^2-\frac{1}{3}\times6^3\right)-(0)=0$ $\bullet^4$ X

$\Rightarrow$ Area $=0\times300=0\ \text{m}^2$ $\bullet^5$ ✓



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| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| 7 | b | | | |
| $\bullet^6$ | ss | set derivative to 2 | $\bullet^6$ $6-2x=2$ | |
| $\bullet^7$ | pd | find point of contact | $\bullet^7$ $x=2,\ y=8$ | |
| $\bullet^8$ | pd | find equation of road | $\bullet^8$ $y=2x+4$ | |
| $\bullet^9$ | ss | find correct integral | $\bullet^9$ $\left[\left(x^2+4x\right)-\left(3x^2-\frac{1}{3}x^3\right)\right]_0^2$ | |
| $\bullet^{10}$ | ic | calculate area | $\bullet^{10}$ $800\text{m}^2$ | **5** |

**Notes:**

6. For candidates who omit '$\text{m}^2$' at both $\bullet^5$ and $\bullet^{10}$ stages, $\bullet^{10}$ is not available.
7. Candidates who arrive at an incorrect equation at $\bullet^8$, or produce an equation ex nihilo, must use an equation of the form $y=2x+c$ with $c>0$, for $\bullet^9$ and $\bullet^{10}$ to be available.
8. $y=2x+4$ must appear explicitly or as part of the integrand for $\bullet^8$ to be awarded.
9. $\bullet^{10}$ is only available as a result of a valid strategy at the $\bullet^9$ stage, ie $\int(\text{line})-(\text{quadratic})$ **and** lower limit = 0 and upper limit < 3.

**Commonly Observed Responses:**

**Candidate D: Alternative Method**

Line has equation of the form $y=2x+c$, $y=2x+c$ and $y=6x-x^2$

intersect where $x^2-4x+c=0$ $\bullet^6$ ✓

tangency $\Rightarrow$ 1 point of intersection

$\Rightarrow b^2-4ac=0$ $\bullet^7$ ✓

$16-4c=0$

$c=4$ $\bullet^8$ ✓

Continue as above.

| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| **8** | | | | |
| $\bullet^1$ | pd | correct values | $\bullet^1$ $g=-p$, $f=-2p$, $c=3p+2$ | |
| $\bullet^2$ | ss | substitute and rearrange | $\bullet^2$ $5p^2-3p-2$ | |
| $\bullet^3$ | ic | knowing condition | $\bullet^3$ $g^2+f^2-c>0$ | |
| $\bullet^4$ | pd | factorise and solve | $\bullet^4$ $(5p+2)(p-1)=0 \Rightarrow p=-\frac{2}{5}, p=1$ | |
| $\bullet^5$ | ic | correct range | $\bullet^5$ $p<-\frac{2}{5}$, $p>1$ | **5** |

**Notes:**

1. Candidates who state the coordinates of the centre, $(p,2p)$ and state the radius, $r=\sqrt{...-(3p+2)}$ gain $\bullet^1$.
2. Accept $(-p)^2+(-2p)^2-(3p+2)$ or $p^2+(2p)^2-(3p+2)$. If brackets are omitted $\bullet^1$ may only be awarded if subsequent working is correct.
3. Do not accept $(-p)^2+(2p)^2-(3p+2)$ or $(p)^2+(-2p)^2-(3p+2)$ for $\bullet^1$.
4. Do not accept $g^2+f^2-c\geq 0$ for $\bullet^3$.
5. For a candidate who uses $c=2$ and follows through to get $p<-\sqrt{\frac{2}{5}}$, $p>\sqrt{\frac{2}{5}}$, award $\bullet^2$, $\bullet^3$ and $\bullet^5$.
6. Evidence for $\bullet^3$ may appear at $\bullet^5$ stage.
7. $\bullet^4$ and $\bullet^5$ can only be awarded for solving a quadratic inequation.

**Commonly Observed Responses:**

**Candidate A**

$g=-2p$, $f=-4p$, c=3p+2 $\bullet^1$ X

$20p^2-3p-2$ $\bullet^2$ ✓

$g^2+f^2-c>0$ $\bullet^3$ ✓

$(4p+1)(5p-2)=0 \Rightarrow p=-\frac{1}{4}, p=\frac{2}{5}$ $\bullet^4$ ✓

$p<-\frac{1}{4}$, $p>\frac{2}{5}$ $\bullet^5$ ✓

**Candidate B**

$(x-p)^2-p^2+(y-2p)^2-4p^2+3p+2=0$

$(x-p)^2+(y-2p)^2$ $\bullet^1$ ✓

$=5p^2-3p-2$ $\bullet^2$ ✓

$5p^2-3p-2>0$ $\bullet^3$ ✓

$(5p+2)(p-1)>0$ $\bullet^4$ ✓

$p<-\frac{2}{5}$, $p>1$ $\bullet^5$ ✓



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| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| 9 | a | | | |
| $\bullet^1$ ss know to differentiate<br>$\bullet^2$ pd differentiates trig. function<br>$\bullet^3$ pd applies chain rule | | | $\bullet^1$ $a = v'(t)$<br>$\bullet^2$ $-8\sin\left(2t-\frac{\pi}{2}\right)\ldots\ldots$<br>$\bullet^3$ $\ldots\ldots\times 2$ and complete $a(t) = -16\sin\left(2t-\frac{\pi}{2}\right)$ | 3 |

**Commonly Observed Responses:**

**Candidate A: Alternative Method**

Part (a)

$v(t) = 8\cos\left(2t-\frac{\pi}{2}\right) = 8\sin 2t$

$v'(t) = \ldots.$ $\bullet^1$ ✓

$= 8\cos 2t\ldots$ $\bullet^2$ ✓

$= \ldots\ldots\times 2$ $\bullet^3$ ✓

Part (b)

$v'(10) = 16\cos 20 = 6\cdot 53$ $\bullet^4$ ✓

$> 0, \Rightarrow$ velocity is increasing $\bullet^5$ ✓

Part (c)

$s(t) = \int v(t)dt$ $\bullet^6$ ✓

$s(t) = -4\cos 2t + c$ $\bullet^7$ ✓

$4 = -4 + c \Rightarrow c = 8$

$\Rightarrow s(t) = -4\cos 2t + 8$ $\bullet^8$ ✓

or $\Rightarrow s(t) = 8 - 4\cos 2t$

**Candidate B:** Candidates who misinterpret the process for rate of change.

Part (a)

$a(t) = \int 8\cos\left(2t-\frac{\pi}{2}\right)dt$

$= 4\sin\left(2t-\frac{\pi}{2}\right)+c$

Wrong process award $0/3$

Part (b)

If $t = 10$, $a = 4\sin\left(20-\frac{\pi}{2}\right)+c$

$= -1\cdot 63 + c$

Cannot evaluate award $0/2$

Part (c)

$s = v'(t)$

$s(t) = -16\sin\left(2t-\frac{\pi}{2}\right)$

Award $2/3$

**Candidate C**

Part (a)

$a = v'(t)$ or equivalent $\bullet^1$

$a = 4\sin\left(2t-\frac{\pi}{2}\right)$ $\bullet^2$ X $\bullet^3$ X

Part (b)

$a(10) = 4\sin\left(20-\frac{\pi}{2}\right) = -1\cdot 63$ $\bullet^4$

<0, So decreasing $\bullet^5$ ✓

Only as a consequence of $\bullet^1$ in part (a)



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| Question | | Generic Scheme | Illustrative Scheme | Max Mark |
|---|---|---|---|---|
| **9** | **b** | | | |
| $\bullet^4$ | ss | know to and evaluate $a(10)$ | $\bullet^4$ $a(10)=6\cdot53$ | |
| $\bullet^5$ | ic | interpret result | $\bullet^5$ $a(10)>0$ therefore increasing | **2** |

**Notes:**

1. $\bullet^5$ is available only as a consequence of substituting into a derivative.
2. $\bullet^4$ and $\bullet^5$ are not available to candidates who work in degrees.
3. $\bullet^2$ and $\bullet^3$ may be awarded if they appear in the working for 9(b). However, $\bullet^1$ requires a clear link between acceleration and $v'(t)$.

| | | | | |
|---|---|---|---|---|
| **9** | **c** | | | |
| $\bullet^6$ | ic | know to integrate | $\bullet^6$ $s(t)=\int v(t)\,dt$ | |
| $\bullet^7$ | pd | integrate correctly | $\bullet^7$ $s(t)=4\sin\left(2t-\frac{\pi}{2}\right)+c$ | |
| $\bullet^8$ | ic | determine constant and complete | $\bullet^8$ $c=8\text{ so }s(t)=4\sin\left(2t-\frac{\pi}{2}\right)+8$ | **3** |

**Notes:**

4. $\bullet^7$ and $\bullet^8$ are not available to candidates who work in degrees. However, accept $\int 8\cos(2t-90)\,dt$ for $\bullet^6$.

[END OF MARKING INSTRUCTIONS]



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